

Anaerobic Co-digestion of Food Waste and Glycerol in Single-Stage System

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Abstract

Resumo

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Chapter 1

Introduction

The generation of organic waste has become one of the defining environmental challenges of our time, particularly as urbanization and industrial activity continue to expand. A major contributor is food waste, a global stream that in the European Union alone is estimated at 88 million tonnes per year, excluding inedible parts removed from the supply chain (Scherhaufer et al., 2018). When considering only post-processing activities – wholesale, retail, food service and households – approximately 66.3 million tonnes of food waste are produced annually in the EU-27+1 (Albizzati et al., 2021). The scale of this loss highlights the urgent need for improved management strategies that move beyond simple disposal.

The environmental burden of food waste is substantial. It accounts for roughly 15–16% of the total climate impact of the entire food supply chain, translating to about 186 Mt CO₂-eq of greenhouse gas emissions, alongside significant contributions to acidification and eutrophication (Scherhaufer et al., 2018). The majority of these impacts arise during primary production, making prevention the single most effective strategy; however, when waste cannot be avoided, its valorisation through anaerobic digestion or composting is favoured over landfill disposal (Albizzati et al., 2021; Scherhaufer et al., 2018). Despite this, much food waste continues to be landfilled or incinerated, forfeiting the opportunity to recover its embedded energy and nutrients.

Parallel to the food waste challenge, the rapid expansion of biodiesel production has generated an excess of another organic by-product: crude glycerol. Formed in the transesterification of triglycerides, approximately 10 kg of crude glycerol are produced for every 100 kg of biodiesel (Garlapati et al., 2016). Global biodiesel output, driven by renewable energy mandates, has caused the glycerol market to swell – with crude glycerol supply projected to reach about 6.33 million tonnes by 2025, of which nearly 680 000 tonnes are expected to be highly impure due to the shift towards waste-based feedstocks (Attarbachhi et al., 2023).

Crude glycerol composition varies widely depending on the oil source and catalyst used,

but it typically contains methanol, soaps, free fatty acids, salts and other matter organic non-glycerol (MONG) compounds (Anand & Saxena, 2012; Attarbachhi et al., 2023). These impurities are known to inhibit microbial growth and fermentation, as demonstrated by the severe reduction in 1,3-propanediol production when untreated crude glycerol from jatropha, soybean or sunflower oils was used (Anand & Saxena, 2012). Purification to technical or pharmaceutical grade is energy-intensive and often uneconomical for small and medium biodiesel plants (Attarbachhi et al., 2023), prompting the search for direct, cost-effective valorisation routes. Because of its high organic content and ready biodegradability, crude glycerol has attracted attention as an ideal co-substrate for anaerobic digestion, where it can serve as a concentrated carbon source while being simultaneously detoxified by dilution and microbial synergy (Gebreegziabher et al., 2025; Tomczak et al., 2025).

Anaerobic digestion (AD) is a mature biological process in which consortia of microorganisms convert organic matter into biogas – primarily methane (55–65%) and carbon dioxide – under oxygen-free conditions, passing through the sequential stages of hydrolysis, acidogenesis, acetogenesis and methanogenesis (Gebreegziabher et al., 2025). AD offers a unique combination of benefits: it produces renewable, storable energy that can replace fossil fuels; generates a nutrient-rich digestate that can be used as organic fertiliser; and reduces the environmental load of organic wastes by avoiding landfilling and its associated methane emissions (Gebreegziabher et al., 2025; Scherhauser et al., 2018).

Within this framework, anaerobic co-digestion – the simultaneous treatment of two or more substrates – has emerged as a particularly effective strategy. Co-digesting crude glycerol with nitrogen-rich materials such as animal manure, sewage sludge or food waste balances the carbon-to-nitrogen ratio, dilutes potential inhibitors, provides essential macro- and micronutrients, and fosters synergistic microbial interactions (Gebreegziabher et al., 2025; Tomczak et al., 2025). The result can be a dramatic improvement in biogas and methane yield: studies report increases ranging from 14% to 226% when crude glycerol is added to a variety of base substrates, with methane contents generally exceeding 60% as long as glycerol does not exceed about 8% (v/v) of the feed (Gebreegziabher et al., 2025; Tomczak et al., 2025).

However, excessive glycerol loads can lead to volatile fatty acid accumulation, pH drop and methanogen inhibition, underscoring the need for careful optimisation (Gebreegziabher et al., 2025; Tomczak et al., 2025). Overall, the convergence of abundant food waste and surplus crude glycerol supplies, together with the proven capability of anaerobic digestion to convert these streams into renewable methane and fertiliser, represents a tangible route towards a circular, low-carbon bioeconomy.

Chapter 2

Objectives

2.1 General Objectives

The primary objective of this dissertation project is to evaluate the efficiency of a single-stage anaerobic co-digestion system using organic food waste and glycerol as substrates. The investigation will be conducted by determining biogas production under different operational conditions, with a focus on industrial feasibility.

The central hypothesis is that the controlled addition of glycerol, a by-product of the biodiesel industry, enhances biogas production due to its high energy content and rapid biodegradability, provided that adequate operational conditions are maintained to avoid volatile fatty acid overload.

2.2 Specific Objectives

1. Characterize the substrates (food waste and glycerol), evaluating physicochemical parameters relevant to anaerobic digestion.
2. Operate the single-stage anaerobic co-digestion system under different operational conditions, assessing process stability.
3. Determine biogas production as a function of substrate proportions and operational conditions.
4. Evaluate the techno-economic feasibility of using glycerol as a co-substrate, considering energy yield and potential for industrial application.

Chapter 3

Literature Review

Chapter 4

Materials and Methods

4.1 Collection and Preparation of the Food Waste

Food waste was collected from the canteen at the Instituto Politécnico de Bragança (IPB), prioritizing residues with high proportions of rice, pasta, and bread to maximize the sugar content and minimize the nitrogen content. The waste was diluted with approximately 2.5 L of distilled water and homogenised using a blender to obtain a liquid paste. It was then stored in a refrigerator at 4 °C to prevent spoilage and suppress microbial and fungal activity (Degueurce et al., 2020).

4.2 Biodiesel Glycerol Collection

A 6 L bottle of crude biodiesel glycerol, obtained from previous biodiesel production experiments at IPB, was provided for this study. Crude glycerol from biodiesel typically contains impurities such as catalyst residues (Attarbach et al., 2023), which may influence its behaviour in anaerobic digestion.

4.3 Inoculum Collection

The anaerobic inoculum consisted of sludge collected from Águas do Norte wastewater treatment plant in Bragança. This ensured the availability of a microbial population for the anaerobic digestion (Posmanik et al., 2020).

4.4 Physical and Chemical Characterisation

The total solids (TS), volatile solids (VS), fixed solids (FS), and density were determined for the food waste, biodiesel glycerol, and inoculum. Total organic carbon (TOC) and total nitrogen (TN) were determined for the food waste and biodiesel glycerol.

4.4.1 TS and VS Determination

TS and VS were measured using a sequential gravimetric method based on Standard Methods 2540 B and 2540 E (Baird et al., 2017). The method involves controlled drying and calcination steps to quantify total solids and volatile solids fractions.

Initially, porcelain crucibles were placed in a muffle furnace at 550 °C for 1 h. For cooling, they were transferred to an oven at 105 °C for 15 min and then moved to a desiccator until they reached room temperature; their empty masses were recorded. For each material, a sample was placed into three crucibles and the wet mass was measured. The crucibles were then dried in an oven at 105 °C for 2 h (biodiesel glycerol required 6–8 h to reach constant mass). After drying, the crucibles were cooled in a desiccator to room temperature and weighed again to determine TS.

The same crucibles were subsequently placed in a muffle furnace at 550 °C for 50 min. Afterwards they were transferred to an oven at 105 °C for 15 min and then moved to a desiccator until they reached room temperature; the final masses were recorded to calculate VS.

TS, VS and FS were calculated as follows:

$$TS = \frac{M_{105} - M_{crucible} * 1000}{V_{sample}} \quad (4.1)$$

$$VS = \frac{M_{105} - M_{550} * 1000}{V_{sample}} \quad (4.2)$$

$$FS = \frac{M_{550} - M_{crucible} * 1000}{V_{sample}} \quad (4.3)$$

where M_{wet} is the mass of crucible plus wet sample, M_{105} is the mass after drying at 105 °C, M_{550} is the mass after ignition at 550 °C, $M_{crucible}$ is the mass of the empty crucible and V_{sample} is the volume of the sample. The volume is in mL and the weight is in mg .

4.4.2 Density Determination

Density was determined by a gravimetric–volumetric method using graduated cylinders. Dry cylinders were weighed empty, then filled with distilled water at 20 °C and weighed

again to verify the reading accuracy (density of water at 20 °C = 998.21 g L⁻¹). After drying, each cylinder was filled with a known volume of sample and weighed. The density of each material was calculated using

$$\rho = \frac{M_{\text{total}} - M_{\text{cylinder}}}{V_{\text{sample}}} \quad (4.4)$$

where M_{total} is the mass of cylinder plus sample, M_{cylinder} is the empty cylinder mass, and V_{sample} is the measured volume.

4.4.3 TOC and TN Determination

Total organic carbon (TOC) and total nitrogen (TN) were measured using a Shimadzu TOC-5000A analyser. Prior to analysis, the food waste was diluted 400-fold and the biodiesel glycerol was diluted 800-fold to bring the concentrations within the linear range of the instrument.

4.5 Experiment Configuration

The reactor had a useful volume of 1 L and was operated as a continuous stirred-tank reactor (CSTR) under anaerobic conditions. Mixing was provided by a magnetic stir bar, and the overflow exit tube was kept closed with a valve to maintain an anaerobic environment.

The reactor was kept at 35 °C, close to the optimal temperature for mesophilic microorganisms (Choorit & Wisarnwan, 2007). The organic loading rate (OLR) was set with a max of 4 kg VS m⁻³ d⁻¹, with a hydraulic retention time (HRT) of 30 days, conditions for stable co-digestion of food waste and glycerol (Tomczak et al., 2025). The pH was maintained around 7 by adding KHCO₃ to the reactor whenever the pH dropped below 7, and by incorporating KHCO₃ into the feed solution at a concentration of 10 g L⁻¹.

The organic loading rate (OLR) and hydraulic retention time (HRT):

$$\text{HRT} = \frac{V}{Q} \quad [\text{d}] \quad (4.5)$$

$$\text{OLR} = \frac{S_{\text{in}} \times Q}{V} \quad [\text{kg VS m}^{-3} \text{ d}^{-1}] \quad (4.6)$$

where S_{in} is the influent VS concentration (kg m⁻³), Q is the volumetric flow rate (m³ d⁻¹), and V is the reactor working volume (m³).

4.5.1 Feeding Solution

The feed was designed to achieve a carbon-to-nitrogen (C/N) ratio of 25:1, which is optimal for anaerobic digestion (Li et al., 2011). Characterisation of the food waste and biodiesel glycerol showed that both substrates contained nitrogen, making it difficult to balance the C/N ratio using both of them together. Therefore, pure glycerol was used as an additive to allow precise control of the C/N ratio. Based on the chosen HRT and the daily volumetric flow rate, the reactor was fed once per day to ensure a flow rate high enough to transport the solid food-waste suspension through the feeding tubes.

Chapter 5

Results

Chapter 6

Discussion

Chapter 7

Conclusion

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